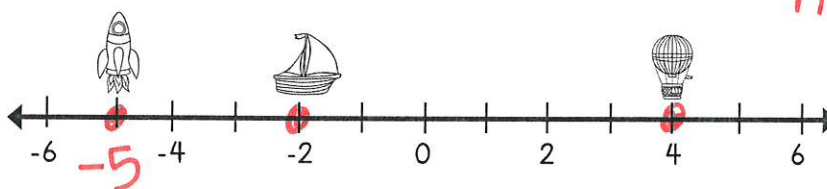
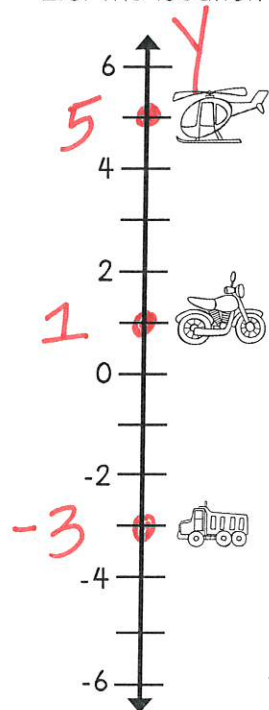


# GRAPHING ON THE COORDINATE PLANE

List the location of the objects on the number lines below.



|                                                     |                                                           |                                                     |
|-----------------------------------------------------|-----------------------------------------------------------|-----------------------------------------------------|
| <b>-3 ON THE VERTICAL NUMBER LINE</b><br>dump truck | <b>4 ON THE HORIZONTAL NUMBER LINE</b><br>hot air balloon | <b>SAILBOAT</b><br>-2 on the horizontal number line |
| <b>MOTORCYCLE</b><br>1 on the vertical number line  | <b>5 ON THE VERTICAL NUMBER LINE</b><br>helicopter        | <b>ROCKET</b><br>-5 on the horizontal number line   |

The coordinate plane extends the two number lines to include integers.

They intersect at a point called the origin, which is located at (0,0).

Label the different parts of the coordinate plane.

**$(-, +)$**

Quadrant II

- x-coordinate

+ y-coordinate

**$(+, +)$**

Quadrant I

+ x-coordinate

+ y-coordinate

**$(-, -)$**

Quadrant III

- x-coordinate

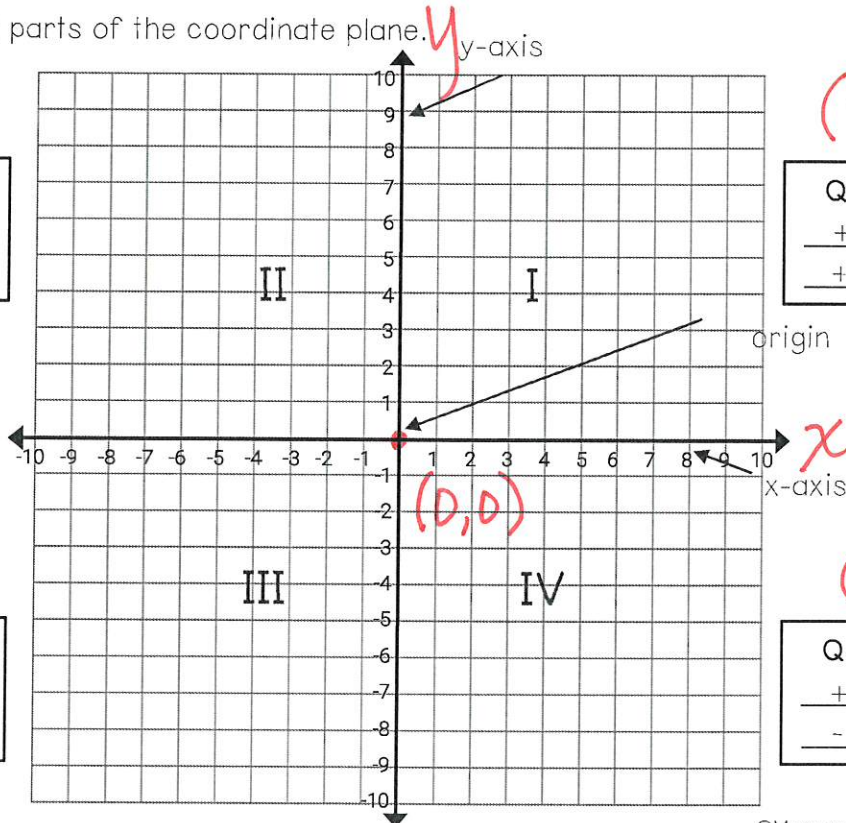
- y-coordinate

**$(+, -)$**

Quadrant IV

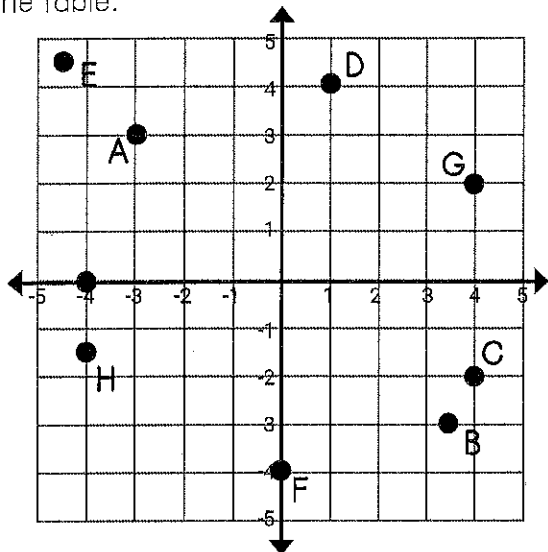
+ x-coordinate

- y-coordinate



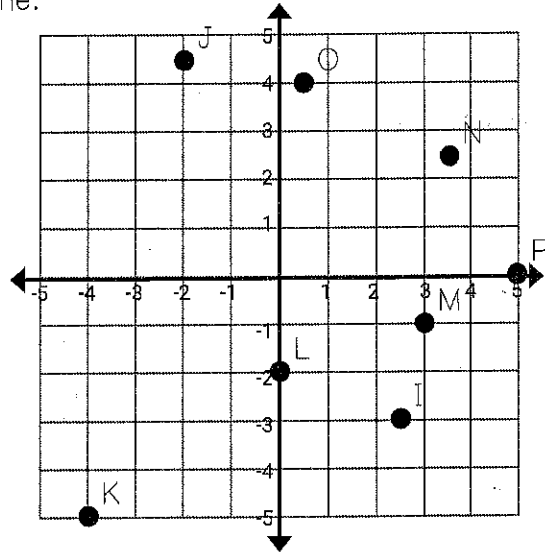
Use the coordinate plane below to answer the questions.

1. Determine the ordered pair for each point in the table.



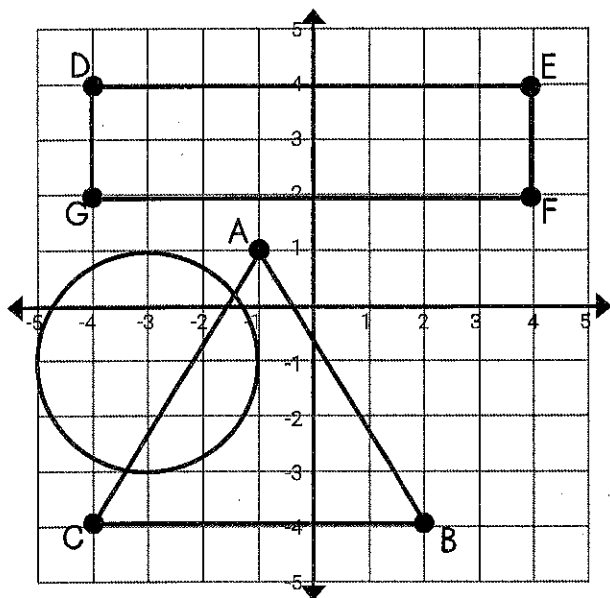
|          |                                 |          |                       |
|----------|---------------------------------|----------|-----------------------|
| <b>A</b> | $(-3, 3)$                       | <b>B</b> | $(3\frac{1}{2}, -3)$  |
| <b>C</b> | $(4, -2)$                       | <b>d</b> | $(1, 4)$              |
| <b>e</b> | $(-4\frac{1}{2}, 4\frac{1}{2})$ | <b>f</b> | $(0, -4)$             |
| <b>g</b> | $(4, 2)$                        | <b>H</b> | $(-4, -1\frac{1}{2})$ |

2. Plot the points below on the coordinate plane.



|          |                    |          |                      |
|----------|--------------------|----------|----------------------|
| <b>i</b> | $(2.5, -3)$        | <b>J</b> | $(-2, 4\frac{1}{2})$ |
| <b>k</b> | $(-4, -5)$         | <b>L</b> | $(0, -2)$            |
| <b>M</b> | $(3, -1)$          | <b>N</b> | $(3.5, 2.5)$         |
| <b>O</b> | $(\frac{1}{2}, 4)$ | <b>P</b> | $(5, 0)$             |

3. The graph below shows several figures. Use the information to answer the questions.



a. Which points meet the conditions where  $x < 1$  and  $y > -1$ ?

A, G, D

b. At approximately which points do the triangle and the circle intersect?

$(-3\frac{1}{2}, -3)$  and  $(-1\frac{1}{2}, \frac{1}{4})$

c. Name the ordered pair that represents the center of the circle.

$(-3, -1)$

Summarize today's lesson: